

CEBU INSTITUTE OF TECHNOLOGY – UNIVERSITY

College of Computer Studies • Department of Information Technology

TASK FORCE BRUNO: INTEGRATED CAMPUS PET MANAGEMENT SYSTEM

MVP Validation Framework & Empirical Research Plan (Capstone 2)

Attribute	Specification Details
Institution	Cebu Institute of Technology – University (CIT-U)
Course Track	Capstone 2 / Information Technology
Document Version	Version 1.0 (Production Validation Baseline)
Target Sample Size	N = 30 Stratified Campus Participants (22 Community, 8 Staff)
Evaluation Frameworks	Technology Acceptance Model (TAM) + ISO 9241-11 Usability + SUS
System Deployment	Vercel (Client) + Render (DRF Gateway) + Supabase (PostgreSQL & Storage)

1. Executive Summary & Validation Goals

The primary objective of the Task Force Bruno MVP Validation is to systematically evaluate the usefulness, usability, operational efficiency, user experience, and institutional acceptance of the deployed campus pet management platform at Cebu Institute of Technology – University (CIT-U).

This validation study gathers empirical evidence across four foundational pillars:

- 1. Stray Identification:** Mitigates identification delays and streamlines animal welfare operations.
- 2. Rapid Lookups:** Dual-card optical QR scanner and AI-driven trait search engine enable fast lookups.
- 3. Clinical Tracking:** Digitized clinical timeline and vaccination journal improve institutional records.
- 4. Institutional Acceptance:** Verifies community willingness to adopt and sustain the system long-term.

2. Research Objectives & Evaluation Matrix

The evaluation connects Capstone research questions with established scientific frameworks and measurable metrics:

RQ / Dimension	Model / Framework	Target Benchmark Metric	Data Instrument
RQ1: Usability	ISO 9241-11 & SUS	SUS Score $\geq 75/100$; PEOU $\geq 4.0/5.0$	10-Item SUS Questionnaire
RQ2: Effectiveness	Goal-Question-Metric	$\geq 90\%$ Task Completion, 0 fatal errors	Moderated Task Protocol Sheet
RQ3: Efficiency	ISO 9241-11 Time-on-Task	Mean QR scan $\leq 25s$; Sighting log $\leq 60s$	Stopwatch / Event Timestamps
RQ4: Usefulness	TAM (Perceived Usefulness)	PU Mean Score $\geq 4.2 / 5.0$	7-Point TAM Survey Instrument
RQ5: User Acceptance	TAM (Behavioral Intent)	$\geq 85\%$ Positive Adoption Intent	TAM Survey + Qualitative Interviews
RQ6: UX & Reliability	UAT & Heuristic Review	Zero blocking UI bugs; $> 80\%$ satisfaction	Post-Study Feedback Survey

3. Selected Validation Models & Frameworks

A Multi-Method Hybrid Approach is adopted to ensure rigorous triangulated findings:

- **TAM Model:** Evaluates Perceived Usefulness (PU), Perceived Ease of Use (PEOU), Attitude (ATU), and Behavioral Intention (BI).
- **ISO 9241-11 Usability:** Measures Effectiveness (task completion), Efficiency (time-on-task), and User Satisfaction.
- **System Usability Scale (SUS):** Standardized 10-item questionnaire providing an industry benchmark score (0-100).
- **Performance Metrics:** Captures objective Task Success Rate (%) and Time-on-Task (seconds).
- **Qualitative Interviews:** Semi-structured post-test questions uncovering UX bottlenecks and user feedback.

4. Target Respondents & Stratified Sampling Strategy

A total sample of N = 30 stakeholders from CIT-U will participate in the validation study:

Cohort Group	Target Sub-Group	Size (n)	Role & Responsibilities in Testing
Cohort A: Community	College Students (All Depts)	14	Test QR scanning, AI search, sightings, adoption
Cohort A: Community	Faculty Members	4	Test campus bulletins, animal identification, sighting alerts
Cohort A: Community	Non-Teaching Staff	4	Test mobile scanning, reporting lost/injured campus pets
Cohort B: Staff / Ops	MDC Staff Members	3	Test administrative triage, health records, bulletins
Cohort B: Staff / Ops	Campus Clinic / Vet Handlers	2	Test clinical logging, anti-rabies vaccine logs & due dates
Cohort B: Staff / Ops	Animal Org Student Officers	3	Test adoption reviews, rescue intake, inventory supply logs

5. Validation Test Scenarios & Task Protocols

Cohort A: Campus Community User Test Protocol (Tasks 1–5)

Task ID	Task Name	Scenario Prompt	Success Criteria	Max Time
TS-A1	Registration & Login	Register with @cit.edu email and ID (XX-XXXX-XXX), then log in.	Profile created, session active, landed on dashboard	60s
TS-A2	Collar QR Lookup	Scan sample pet QR tag (PET-0001) to retrieve animal file.	jsQR decodes matrix, loads demographics and vaccine logs	45s
TS-A3	AI Trait Search	Search for "calico coat with white socks near library".	Candidate profiles matched descending by affinity score	30s
TS-A4	Submit Sighting	Report stray sighting near Gym with description and photo.	Sighting saved to Supabase with image and location	75s
TS-A5	Adoption Form	Browse gallery, select companion, submit adoption form.	Application stored with status set to Pending	90s

Cohort B:
MDC Staff
& Clinic Admin
Test Protocol (Tasks

Task ID	Task Name	Scenario Prompt	Success Criteria	Max Time
TS-B1	New Animal Intake	Register rescued stray with biometrics, TNR tag, and image upload.	Pet record created, visual asset saved in Supabase bucket	120s
TS-B2	Sighting Triage	Review pending sighting alert, verify evidence, and mark as Triaged.	Sighting status updated in database in real time	45s
TS-B3	Clinical & Vaccine Log	Log veterinary diagnosis and vaccine with calculated due date.	Record and vaccine journal appended to health timeline	60s
TS-B4	Adoption Review	Inspect pending application, verify applicant, approve adoption.	Application marked Approved, pet moved to Adopted Alumni	60s
TS-B5	Inventory Ledger	Log inbound donation of 10 cat food units with donor remarks.	Inventory level updated, transaction audit ledger appended	45s

6. Data Collection Instruments & Questionnaires

Instrument 1: System Usability Scale (SUS) (5-Point Likert Scale)

1. I think that I would like to use Task Force Bruno frequently.

2. I found the system unnecessarily complex.

3. I thought the system was easy to use.

4. I think that I would need the support of a technical person to use this system.

5. I found the various functions in this system were well integrated.

6. I thought there was too much inconsistency in this system.

7. I would imagine that most people would learn

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quickly.

8. I
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Q2)+(
Q3-1)+
(5-Q4)
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)+(5-Q
6)+(Q7
-1)+(5-
Q8)+(
Q9-1)+
(5-
Q10)]
* 2.5

Instrument 2: Technology Acceptance Model (TAM) Items (7-Point Likert)

- **PU1-PU4 (Perceived Usefulness):** Speed of identification, transparency of vaccine/TNR logs, community engagement, and overall campus utility.
- **PEOU1-PEOU4 (Ease of Use):** Ease of QR scanning, clarity of navigation, ease of form submission, and overall UI ergonomics.
- **ATU1-ATU2 (Attitude):** Positive sentiment toward digital animal welfare vs legacy paper records.
- **BI1-BI2 (Behavioral Intent):** Intention to report sightings and recommend the system to peers.

7. Measurable Success Criteria & Target Benchmarks

Evaluation Parameter	Minimum Threshold	Target Benchmark	Operational Interpretation
System Usability Scale (SUS)	>= 68.0 (Global Avg)	>= 75.0 (Grade A - Good to Excellent)	Superior usability with minimal friction.
Task Success Rate (TSR)	>= 80.0%	>= 90.0% Completion Rate	Users can independently complete key workflows.
Mean Time (QR Scan)	<= 60 seconds	<= 25 seconds	Collar decoding is swift in field sightings.
Mean Time (Sighting Log)	<= 120 seconds	<= 60 seconds	Stray reporting friction is very low.
TAM: Perceived Usefulness	>= 3.5 / 5.0	>= 4.2 / 5.0 (Positive)	Strong validation of animal welfare utility.
TAM: Ease of Use	>= 3.5 / 5.0	>= 4.0 / 5.0 (Positive)	High learnability for non-technical users.
TAM: Behavioral Intention	>= 70.0% Positive	>= 85.0% Positive Intent	High likelihood of recurring adoption.

8. Data Analysis Methodology

- Descriptive Statistics: Mean, SD, and Median values for all SUS and TAM constructs.
- Reliability Testing: Cronbach's Alpha (target alpha >= 0.70) for TAM scale consistency.
- Correlation Analysis: Pearson's r assessing PEOU -> PU -> Behavioral Intention.
- Thematic Coding: Inductive synthesis of qualitative user interview transcripts.

9. Validation Execution Roadmap

Phase	Timeline	Key Activities	Deliverables
Phase 1: Prep	Days 1–3	Finalize instruments, deploy cloud build, seed QR tags	Printed/Digital Survey Forms & Live App
Phase 2: Testing	Days 4–8	Conduct 30 moderated testing sessions (22 user, 8 staff)	Completed Sheets, SUS/TAM Data, Notes
Phase 3: Analysis	Days 9–11	Compute SUS/TAM aggregates, Task Success & ToT	Statistical Tables, Charts & Findings
Phase 4: Reporting	Days 12–14	Draft Capstone Chapter 4, implement UX fixes	Completed Capstone Chapter 4 & Slide Deck

10. Adviser Consultation & Sign-Off Section

- [] Validation Model Approval: Concurrence on hybrid TAM + ISO 9241-11 + SUS methodology.
- [] Respondent Stratification: Approval of N = 30 cohort distribution (22 Community, 8 Staff).
- [] Instrument Review: Verification of 10-item SUS and 12-item TAM survey questionnaires.
- [] Data Privacy Compliance: Assured anonymous handling of participant data under CIT-U ethical guidelines.
- [] Live Deployment Verification: Production cloud build operational on Vercel and Render.

Role	Name & Designation	Signature	Date of Approval
Capstone Adviser	_____	_____	___ / ___ / 2026
Department Chair	_____	_____	___ / ___ / 2026
Lead Researcher	Clark Yu (Task Force Bruno)	_____	___ / ___ / 2026

